

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 150,000 over 24 months is declined under the default policy (v1.0). Verified monthly income is Rs 21,750 against existing obligations of Rs 6,000; the proposed EMI of Rs 7,344 gives a FOIR of 61.4% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: MEDIUM. Reason codes: HIGH_FOIR.

For the borrower:

Arjun Mehta, we could not approve the loan of Rs 150,000 as requested because the monthly instalment of Rs 7,344 would take 61.4% of your verified income (our limit is 50.0%). The smallest change that would make it approvable is to extend tenure 24 -> 42 months, which brings the instalment to Rs 4,687 and FOIR to 49.1%.

Key drivers:

- FOIR 61.4% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 21,750 (volatility 4.4%)
- HIGH_FOIR: FOIR 61.35% exceeds policy limit 50.00%

Next steps:

- Offer the borrower the approvable alternative: extend tenure 24 -> 42 months.
- Share the plain-language explanation with the borrower.

1. DECISION

Decision: DECLINED Confidence: 0.82

Reason codes: HIGH_FOIR

- [blocker] HIGH_FOIR: FOIR 61.35% exceeds policy limit 50.00%

1a. PATH TO APPROVAL (automatic what-if on the immutable base run)

Minimum change: extend tenure 24 -> 42 months

change.tenure_months: 42

resulting EMI 4,687.43 FOIR 0.4914 LTV 0.75 -> APPROVED

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Arjun Mehta's loan is declined because a hard policy rule failed: FOIR 61.4% vs limit 50.0% (+11.4 pp over).

The smallest fix is to extend tenure 24 -> 42 months, which would bring FOIR to 49.1% and the EMI to Rs 4,687, turning the outcome into approved.

The model's risk band is MEDIUM (score 0.4524 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: foir (obligations / income) (61.4% vs a typical 41.1%, +0.061); ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); existing monthly obligations (Rs 6,000 vs a typical Rs 9,458, +0.004).

What pulls risk down: employment stability (95.6% vs a typical 71.8%, -0.024); income volatility (4.4% vs a typical 22.4%, -0.018); income mismatch across documents (0.0% vs a typical 12.7%, -0.013).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

FAIL FOIR 61.4% vs limit 50.0% (+11.4 pp over)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

PASS age 26 vs minimum 21 (+5 years)

PASS verified monthly income Rs 21,750

risk score 0.452358 = base 0.437336 + sum SHAP 0.015022 (method additive_exact, additivity error 0.0)

phi(foir) = +0.0608 [value 61.4% vs baseline 41.1%]

phi(employment_stability) = -0.0237 [value 95.6% vs baseline 71.8%]

phi(income_volatility) = -0.0179 [value 4.4% vs baseline 22.4%]

phi(income_mismatch_percentage) = -0.0127 [value 0.0% vs baseline 12.7%]

phi(ltv) = +0.0124 [value 75.0% vs baseline 68.1%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-FOIR-003

full_name: Arjun Mehta
age: 26
location_tier: tier_2
employment_type: gig_platform
declared_monthly_income: 22,000.00
loan_request.amount: 150,000.00
loan_request.down_payment: 50,000.00
loan_request.tenure_months: 24
loan_request.vehicle_price: 200,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Arjun Mehta conf 0.95 [application_form p.None] auto
monthly_income = 22,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 150,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 200,000.00 conf 0.95 [application_form p.None] auto
full_name = Arjun Mehta conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-1122 conf 0.9 [kyc.txt p.1] auto
address = 7 Station Road, Jaipur 302001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 22,000.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [22000.0, 21500.0, 23800.0, 20900.0, 22600.0, 21200.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 6,000.00 conf 0.92 [bank_statement.txt p.1] auto
address = 7 Station Road, Jaipur 302001 conf 0.85 [bank_statement.txt p.1] auto
vehicle_price = 200,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 150,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 22,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 22,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 21,750.00
existing_obligations: 6,000.00
emi: 7,344.47
foir: 0.6135
ltv: 0.75
income_volatility: 0.0443
income_stability: 0.9557
loan_to_income: 0.5747
emi_to_income: 0.3377
input.annual_rate: 16.00
input.existing_monthly_obligations: 6,000.00
input.loan_amount: 150,000.00
input.monthly_incomes: [22000.0, 21500.0, 23800.0, 20900.0, 22600.0, 21200.0]
input.tenure_months: 24
input.vehicle_price: 200,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.4524 band MEDIUM model xgb-v1/fallback

6. AI REASONING

strengths: LTV 75% leaves borrower equity cushion
risks: FOIR 61% strains monthly cash flow
reasoning: Deterministic FOIR 61.35%, LTV 75.00%; XGBoost band MEDIUM (score 0.4524).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.